Power

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Compare how quickly energy is transferred.

SUCCESS CRITERIA - I can...

1. Distinguish energy in J from power in W.
2. Use power=energy/time.
3. Explain equal work done at different rates.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “power” and “work” to explain your answer. Sentence starters are optional.

power: rate of energy transfer

work: energy transferred by a force along a displacement

interval: the time or range between two stated values

01 Warm-up: bring back an earlier idea

A rolling object loses **15 J** gravitational energy but gains only **9 J** translational kinetic energy. **Name two possible destinations of** the difference.

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $P_{avg} = W/\Delta t$ = energy transferred/Δt; P = Fv. Second equation: force parallel to velocity. Power units: **1 W = 1 J/s**.

Worked reasoning: Two motors each transfer **120 J**. One takes **3 s**, the other **6 s**. Their average powers are **40** and **20 W**; total transferred energy is equal.

Your turn: Draw equal energy-transfer arrows and **label the different times** and powers.

Sentence starter: Both ... . They differ in ... because ... .

Case	Energy (J)	Time (s)	Power (W)
1	____	____	____
2	____	____	____

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Two motors each transfer **120 J**. One takes **2 s**; the other **6 s**. Which has greater average power?

Sentence starter: Both ... . They differ in ... because ... .

Motor	Energy	Time
A	120 J	2 s
B	120 J	6 s

Marzano check - circle: ① ② ③

04 Apply the physics

A small motor transfers **90 J** in **5 s**. **Calculate its average power**. If it keeps transferring energy at that rate, **how much** does it transfer in **6 s**? **Complete the comparison table**.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Case	Energy (J)	Time (s)	Power (W)
1	____	____	____
2	____	____	____

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A **100 W** label is described as "**100 joules**." Correct the statement and **give time information** that would make an energy calculation possible.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A cart's kinetic energy changes from **24 J** to **4 J**. **Find net work** and state whether its speed rises or falls. **Show the energy changes** and transfers; **label units**.

Word help: kinetic energy = energy associated with motion

Sentence starter: I found ... [unit]. This value means ... in this situation.

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

07 Exit check

Two students do the same lift. One finishes in half the time. **Compare work** and power.

Sentence starter: Both ... . They differ in ... because ... .

Case	Energy (J)	Time (s)	Power (W)
1	____	____	____
2	____	____	____

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Create two devices with the same **50 W** average power but different energy and time values.

Sentence starter: Both ... . They differ in ... because ... .

Case	Energy (J)	Time (s)	Power (W)
1	____	____	____
2	____	____	____

Marzano check - circle: 1 2 3